

9 CAN Communication

The CAN communication is based on the SAE J1939 protocol. The hardware is supported as in the protocol and messages are using the open addresses 0x18FFiixx and 0x19FFiixx.

ii = Message ID

xx = Note (Unit address)

yy = Master (Host address)

9.1 Standard CAN communication

Standard CAN communication is used to broadcast most important data on the CAN bus
128bit 125kbps ->976msg/s

All values reaching 2 or more bytes are always send with MSB first.

Source	ID- start	ID- end	Message	Rate (Msg/s)	Bus %
	0x18FF00xx		Battery - General Status	1	0.102
	0x18FF01xx		Battery - SOC, SOC time and DOD Ah		
	0x18FF02xx		Battery - Voltages, Currents & Cell min/max		
	0x18FF03xx		Battery - Terminal Voltage, IO1,2,3 Voltages		
	0x18FF05xx		Failure code Buffer	1	0.102
	0x18FF06xx		Cell Temperature	1	0.102
	0x18FF07xx		Production date and Serial number		
	0x18FF10xx		Battery - Cell Voltages		
	0x18FF18xx		Battery - Cell SOC		
	0x19FF00xx		Production date and Serial number	0	
	0x19FF01xx		Firmware update date and Second Serial number	0	
	0x19FF02xx		Firmware and hardware revision	0	
	0x19FF03xx	0x19FF04xx	Clayton Power Part number	0	
	0x19FF05xx	0x19FF08xx	Clayton Power Part Description	0	
	0x19FFF0xx		Test Mode Status	0	
	0x19FFF1xx		Test Variables	0	
Total				5	0.51%

Battery System Voltage

Battery System Current

Battery system Temperature avg

Battery system Temperature min

Battery system Temperature max

9.1.1 0x18FF00xx – Battery - General Status

Reports general status of the DC/DC.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF00xx	Operating State	Failure Level	Controller	Controller Set State	Wakeup Flags		Output Pins	Battery Count

a. Operating State values is a specified in the 0 Operating state diagram.

b. Failure level is a specified in: 4.2 Failure levels

c. Controller of the Operating State

Value	Bit Name	Description
0	No Controller	If no controller is selected will the unit power down after 10 sec
5	I/O Interface	Folia button or wakeup input is the controller
20	SW Master	Receiving master control messages via Single wire
30	CAN Master	Receiving master control messages via CAN
50	Battery Slave	
100	Test Mode	Receiving Test Mode messages via CAN or Single wire

d. Controller requested operating state - same as operating states

e. Li-G4 Wakeup flags

Bit #	Bit Name	Description
0	CAN Activity	Set only OS_READY, Necessary for CAN host to set Controller State to change to OS_CONNECTED
1	SW Activity	Set only OS_READY, Necessary for single Wire host to set Controller State to change to OS_CONNECTED
2	Pre-Charge Voltage	Only present 60sec after OS_CONNECTED - Voltage applied into terminals of the battery must be >= (Battery Voltage - 100mV)
3	Charge Current	Charge Current into the battery
4	Input 1	Voltage on Input 1 above trigger level
5	Input 2	Voltage on Input 2 above trigger level
6	Input 3	Voltage on Input 3 above trigger level
7	Button	Button Wakeup Press
8	G3 Combi Mains connection	Single wire Communication controlled. Message from G3 combi accepting 230vac mains voltage
9	Not Used	
10	Not Used	
11	Not Used	
12	Not Used	
13	Not Used	
14	Not Used	
15	Ignore Wakeups	If this flag is set, will other wakeup flags be ignored. Toggling one of the wakeups will clear this flag.

f. Output Pins

Bit #	Bit Name	Description
0	Wakeup 1 Output	Wakeup 1 pin is active as output
1	Wakeup 2 Output	Wakeup 2 pin is active as output
2	Not Used	
3	Contactor	Main relay inside battery is connected and power can be charged / discharged

g. Battery Count: The number of batteries from on CAN communication used to the Battery System.

h. Transmit period: 1 second

9.1.2 0x18FF01xx - Battery - SOC Status

Reports SOC, time remaining and usable capacity

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF01xx	SOC		SOC time		DOD Ah		0x0000	

a. The SOC of the battery of the usable capacity. Format of value is 1/65536.

b. Time remaining until battery is full (>99%) or empty. Signed value in minutes.

c. DOD Ah is the amount of Ah specified available of the battery.

d. Transmit period: 1 second.

9.1.3 0x18FF02xx – Battery – Voltage and Current

Report battery voltage and current

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF02xx	Battery Voltage		Battery Current		Cell Voltage Min		Cell Voltage Max	

- Battery Voltage is voltage measured on the battery. Resolution is 1mV
- Battery Current is the current going in/or of the battery. Resolution is 100mA (signed)
- Cell Voltage Min is the cell voltage of the lowest cell. Format of value is 1/8192.
- Cell Voltage Max is the cell voltage of the highest cell. Format of value is 1/8192.
- Transmit period: 1 second

9.1.4 0x18FF03xx – Battery – DC Output Voltage and IO Voltages

Report Battery terminal voltage and IO voltages

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF03xx	Battery Terminal Voltage		IO 1 Voltage		IO 2 Voltage		IO 3 Voltage	

- Battery Terminal Voltage is voltage measured on the battery terminals. Resolution is 1mV
- IO x Voltages is the measured voltage on the IO pin. Resolution is 1mV
- Transmit period: 1 second

9.1.5 0x18FF05xx – Battery – Failure Code Buffer

The Failure Code Buffer Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF05xx	FC_0	FC_1	FC_2	FC_3	FC_4	FC_5	FC_6	FC_7

- FC_0 to FC_7 is actual failures in unit. If values are 'Zero' meaning no failure. Failure codes can be found in failure code table.
- Transmit period: 1 second

9.1.6 0x18FF06xx – Internal temperature

Report cell temperature

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF06xx	Cell 1/2 Temperature		Cell 2/3 Temperature		Cell 3/4 Temperature		0x0000	

- Cell Temperature measured between cell 1 & 2.
- Cell Temperature measured between cell 2 & 3.
- Cell Temperature measured between cell 3 & 4.
- Resolution of all temperatures is 1/256 °C/bit (Signed)
- Transmit period: 1s

9.1.7 0x18FF10xx – Battery – Cell Voltages

Report cell voltages

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF10xx	Cell 1 Voltage		Cell 2 Voltage		Cell 3 Voltage		Cell 4 Voltage	

- Cell Voltage is the individual cell voltage. Cell 1 is located at minus. Resolution is 1/8192V
- Transmit period: 1 second

9.1.8 0x18FF18xx – Battery – Cell SOC

Report cell voltages

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF18xx	Cell 1 SOC		Cell 2 SOC		Cell 3 SOC		Cell 4 Soc	

- Cell 1 is the SOC of the first cell (at the negative battery terminal).

- b. Cell SOC shall be unsigned 16 bits, MSB first, with a resolution of 1/65535 % (i.e. 0,00001529%) per bit.
- c. Transmit period: 1 second.

9.1.9 0x18FF0xx – Battery – Master Message

Control message for all connected slaves

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF0xx	Requested operating state	LED Flash sync						

- a. Cell SOC is the individual Cell SOC Calculations. Resolution is 1/8192V
- b. Transmit period: 10 second

9.1.10 0x19FF00xx – Serial Number and Manufacture Date

Reports unit serial number and date of manufacture.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF00xx	Year	Month	Day		Batch Number			Serial Number

- a. Manufacture Date consists of 3 bytes with values from 0-99, 0-12 and 0-31 decimal. Date Format: DD:MM:20YY
- b. In cases where Date is not used, values will be either 0x00 or 0xFF.
- c. Batch Number is number of manufacture batch of BMS. Batch number consists from 3 bytes each can have the value 0-99 which gives a batch number of 6 digits. Format: 000000.
- d. Serial number is the serial number of the BMS. The serial number consists of 2 bytes each can have the value 0-99 which gives a serial number of 4 digits. Format: 0000
The Unit serial number uses both Batch number and Serial number. Format: 000000-0000
- e. Transmit period: After Power up and on request

9.1.11 0x19FF01xx – Second Serial Number and Firmware update date

Reports the second serial number and date for last firmware update.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF01xx	Year	Month	Day		Second Batch Number			Second Serial Number

- a. Last Firmware Update Date consists of 3 bytes with values from 0-99, 0-12 and 0-31 decimal. Date Format: DD:MM:20YY
- b. In cases where Date is not used, values will be either 0x00 or 0xFF.
- c. Batch Number is number of manufacture batch of BMS. Batch number consists from 3 bytes each can have the value 0-99 which gives a batch number of 6 digits. Format: 000000.
- d. Serial number is the serial number of the BMS. The serial number consists of 2 bytes each can have the value 0-99 which gives a serial number of 4 digits. Format: 0000
The Unit serial number uses both Batch number and Serial number. Format: 000000-0000
- e. Transmit period: After Power up and on request

9.1.12 0x19FF02xx – Firmware and Hardware Revision

Reports the software and hardware revisions over CAN.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF02xx	FW Major Revision	FW Minor Revision	FW Build Revision	HW Major Revision	HW Minor Revision	FW2 Major Revision	FW2 Minor Revision	FW2 Build Revision

- FW Revision consists of 3 bytes with each values from 00-99 decimal. Format: 00:00:00
- HW Revision consists of 2 bytes with each values from 00-99 decimal. Format: 00:00
- FW2 Revision consists of 3 bytes with each values from 00-99 decimal. Format: 00:00:00
- Transmit period: After Power up and on request

9.1.13 0x19FF03xx & 0x19FF04xx – Clayton Power Part number

Reports the CP Part number over CAN. The Part number consists of 16 chars.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF03xx	Char n	Char n+1	Char n+2	Char n+3	Char n+4	Char n+5	Char n+6	Char n+7
0x19FF04xx	Char n+8	Char n+9	Char n+10	Char n+11	Char n+12	Char n+13	Char n+14	Char n+15

- Part number is at string (ASCII) of 16 characters.
- If part number is shorter than 16 characters empty bytes will have value '0'
- Transmit period: After Power up and on request

9.1.14 0x19FF05xx - 0x19FF08xx – Clayton Power Part Description

Reports the CP Part Description over CAN. The Part Description consists of 32 chars.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF05xx	Char n	Char n+1	Char n+2	Char n+3	Char n+4	Char n+5	Char n+6	Char n+7
0x19FF06xx	Char n+8	Char n+9	Char n+10	Char n+11	Char n+12	Char n+13	Char n+14	Char n+15
0x19FF07xx	Char n+16	Char n+17	Char n+18	Char n+19	Char n+20	Char n+21	Char n+22	Char n+23
0x19FF08xx	Char n+24	Char n+25	Char n+26	Char n+27	Char n+28	Char n+29	Char n+30	Char n+31

- Part description is at string (ASCII) of 32 characters.
- If part description is shorter than 32 characters empty bytes will have value '0'
- Transmit period: After Power up and on request

9.1.15 0x18EFxxyy Master Control message

The DC/DC converter can be controlled in direction, voltage, current and voltage levels. The Message has to be received periodically for the DC/DC to remain in controlled mode.

Address xx = device address

Address yy = Host address

Request message:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18EFxxyy	Operating State	Operation Mode	Supply Mode	0x00	Output Voltage		Output Current	

- Operating state is the desired state of operation.
- Operation Mode is in controlled mode, mainly for selecting LED for indication.
- Supply mode are used for preset input / Output limits for failure indication.
- Output voltage is used to set level in constant voltage mode. Resolution is 1mV Unsigned integer
- Output voltage is used to set limit in constant current mode. Resolution is 10mA Signed integer
- Refresh rate: <5sec. hereafter it will return to self controlled mode.

9.1.16 0x18EAxxyy Request PGN

Request message:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18EAxxyy	Message ID			0x00	0x00	0x00	0x00	0x00

Device ID = 0x00 (Global address)

Host (PC) = 0xFE

Ex: Request Identification: 0x18EA00FE 01 FF 00 00 00 00 00 00

Ex: Request Test bytes: 0x18EA00FE 01 FF F1 00 00 00 00 00

9.2 CAN Extra Communication

The CAN Extra protocol is used for different applications and functions and these commands are not supported by all applications and should only be used in the right operating modes. The commands not defined in this table are reserved for future applications.

Address xx = device address

Address yy = Host address

Messages received by the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxyyy	0x00	Command						
0x19EFxyyy	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

Messages transmitted from the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFyyxx	0x00	Command						
0x19EFyyxx	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

- Message number: The message number indicates if the message is the handshake or data messages
- Command: see command table
- Data byte 3 to 8 is configured depending of the command.

Commands:

Command	Command Name	Usage	Description
0x00	Acknowledgement Response	All	Response after one way commands
0x01	Enter to Boot mode.	Application	The system enters to boot mode.
0x06	Reboot processor	Application	Reboot processor
0x07	Stop CAN traffic	Application	Stop CAN traffic
0x08	Start CAN traffic	Application	Start CAN traffic
0x0A	Request Settings Transfer	Settings	Request a setting
0x0B	Respond Settings	Settings	Respond the requested setting
0x0C	Receive Settings	Settings	Receive a setting
0x0D	Enter to test mode	Application	Test mode is activated
0x0E	Leave test mode	Application	Test mode is deactivated
0x0F	Load default	Application	
0x10	Request BIOS Settings Transfer	BIOS Settings	Request a BIOS setting
0x11	Respond BIOS Settings	BIOS Settings	Respond the requested BIOS setting
0x12	Receive BIOS Settings	BIOS Settings	Receive a BIOS setting
0x20	Calibration		
0x40	App Settings Request Value	Application	Return Settings value (BlockID, ID)
0x41	App Settings Save Value	Application	Save and Return Settings value (BlockID, ID)
0x42	App Settings Request Default	Application	Return Settings default (BlockID, ID)
0x43	App Settings Request Min	Application	Return Settings min (BlockID, ID)
0x44	App Settings Request Max	Application	Return Settings max (BlockID, ID)

9.2.1 Acknowledgement Response message

After receiving data, calibrations or other similar messages, the unit response with this message to confirm receive and acceptance.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFyyxx	0x00	0x00 Command	Response from Command	Result	Value #0		Value #1	

- Response From Command is the value of the received command what generates the response.
- Result is a enum of different acknowledgements to cover all functions.

Result	Status
-2	Incorrect CalPtr value
-1	Calibration value out of range
0	Unable to save to EEPROM
1	Calibration success

- Value #0 and Value #1 are response values from the different functions. The content of these values depend of the function giving response.

Command	Content of Value #0	Content of Value #1
0x0F	Empty	Empty
0x12	Empty	Empty
0x20	Calibration index pointer	New Calibration scaling value

9.2.2 EEPROM functions

The EEPROM of the Li-G4 can be accessed via CAN communication and settings can be edited. When requesting data, the Li-G4 will return a series of messages containing EEPROM data and a checksum of the data. When sending data to the Li-G4 the checksum of the data has to be calculated before start of transition.

When changing settings in EEPROM, the Li-G4 needs to reboot before changes takes effect. Settings are only read into RAM while booting the micro.

Messages received by the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x0A or 0x0C	EEPROM Address		Byte Count		Checksum of Data (CRC16)	
0x19EFxxyy	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

Messages transmitted from the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFyyxx	0x00	0x0B	EEPROM Address		Byte Count		Checksum of Data (CRC16)	
0x19EFyyxx	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

9.2.3 0x19FFF1xx - Test Variables

Reports the CP Part Description over CAN. The Part Description consists of 32 chars.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FFF1xx	0x00	Test[0]		Test[1]		Test[2]		
0x19FFF1xx	0x01	Test[3]		Test[4]		Test[5]		
0x19FFF1xx	0x02	Test[6]		Test[7]		Test[8]		
0x19FFF1xx	0x03	Test[9]		Test[10]		Test[11]		
0x19FFF1xx	0x04	Test[12]		Test[13]		Test[14]		
0x19FFF1xx	0x05	Test[15]						

- Test variables is a table with 16 unsigned integers send MSB.
- Transmit period: On request

9.2.4 Load default

The Li-G4 Is able to load default settings for both BIOS and Application. The message has to contain a code to avoid random mistakes to overwrite settings.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x0F	Code #0	Code #1	Code #2	Code #3		

- BIOS Settings, when overwriting BIOS settings are all calibrations, serial number and manufacture information lost.
- BIOS Code:

Code #	Value
0	1
1	2
2	3
3	4
- APP Settings, when overwriting APP settings are user settings lost and manually changed settings will return to default. Cell SOC and other self learned settings are also reset.
- App Code:

Code #	Value
0	2
1	3
2	4
3	5

- After any of the loadings of default settings is it necessary to perform a reset of the micro before changes takes effect.

9.2.5 BIOS Settings functions

The BIOS Settings of the Li-G4 can be accessed via CAN communication and settings can be edited. When requesting data, the Li-G4 will return a series of messages containing BIOS Settings data and a checksum of the data. When sending data to the Li-G4 the checksum of the data has to be calculated before start of transition.

When changing settings in BIOS, the Li-G4 needs to reboot before changes takes effect. Settings are only read into RAM while booting the micro.

Messages received by the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x10 or 0x12	BIOS Setting Address		Byte Count		Checksum of Data (CRC16)	
0x19EFxxyy	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

Messages transmitted from the unit:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFyyxx	0x00	0x11	BIOS Settings Address		Byte Count		Checksum of Data (CRC16)	
0x19EFyyxx	Message # (0x01 to 0xFF)	Data n	Data n+1	Data n+2	Data n+3	Data n+4	Data n+5	Data n+6

Address	Byte Count	Direction	Name
3	5	(MSB First)	Serial Number
8	3	(MSB First)	Manufactory Date
11	2	(MSB First)	Hardware Version
13	5	(MSB First)	Serial Number2
18	3	(MSB First)	Software Version2
21	3	(MSB First)	Firmware Date
24	16	(MSB First)	Part Number
40	32	(MSB First)	Part Description

9.2.6 Test mode Functions

The Li-G4 can be placed in test mode. Here is a list of functions available in test mode.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	Command 0x0D or 0x0E	Test Mode Level	Test mode Function	Action			

- a. Command, Test mode can be activated by sending the command: 0x0D and set the test mode level 0,2.

Test mode can be disabled either by sending the command: 0x0E or setting test mode level to '0' as above.

Test mode level must be refreshed, otherwise a timeout will disable test mode

Test Mode Level

Test mode level	Name	Refresh timeout	Description
0	Normal mode		LPS is NOT in test mode and has normal function
2	Test mode	60 sec	Only Test mode functions are active. Most Application functions are not operating in this mode.

- b. Function and Action.

The test mode must be set prior sending a command and same level must be included while sending a command.

Test mode level	Test mode Function	Name	Action	Description
2	1	Remote IO	0 = Off, 1 = On	
	2	Data IO	0 = Off, 1 = On	
	30	Mosfet PSU	0 = Off, 1 = On	Start Power supply on mosfet switch board
	31	Mosfet Charger Enable	0 = Off, 1 = On	Short circuit reset is toggled prior activating. Start charge mosfet
	32	Mosfet Discharge Enable	0 = Off, 1 = On	Short circuit reset is toggled prior activating. Start Discharge mosfet
	33	Pre-Charge Enable	0 = Off, 1 = On	Start Pre-charge function
	50	LED test	0 = Off, 1 = Green On 2 = Red On	Activation of LED's
	51	Buzzer	0 = Off, 1 = On	Start Buzzer
	100	Cell Balance	Bitmap: Cell 1 = Bit 0 Cell 2 = Bit 1 Cell 3 = Bit 2 Cell 4 = Bit 3	Start and stop cell balance for each cell. When bits are 1 is cell balance ON, 0 = Off

9.2.7 Calibration

Calibration of all measurements can be done via CAN communication.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxyy	0x00	0x20	CalPtr	Calibration Value (MSB first)				

a. Calibration of measures.

Calibration values are calculated into a fixed point fractional signed long where the upper 16bit is value and lower 16bit is fragment.

CalPtr	Calibration Name	Data Resolution	Default Value
0	Cell 1 Voltage	1/8192 V (unsigned integer) (Value*8192)	4,94995V ¹
1	Cell 2 Voltage	1/8192 V (unsigned integer) (Value*8192)	4,94995V ¹
2	Cell 3 Voltage	1/8192 V (unsigned integer) (Value*8192)	4,94995V ¹
3	Cell 4 Voltage	1/8192 V (unsigned integer) (Value*8192)	4,94995V ¹
98	Battery Current X200	10mA (Signed Integer) (Value*100)	280,00A
99	Battery Current X20	100mA (Signed Integer) (Value*10)	2800,00A
101	Battery Voltage	1mV (Unsigned Integer) (Value * 1000)	25,50V
107	DC Output Voltage	1mV (Unsigned Integer) (Value * 1000)	63,80V
113	Remote IO Voltage	1mV (Unsigned Integer) (Value * 1000)	55,50V
114	Data IO Voltage	1mV (Unsigned Integer) (Value * 1000)	55,50V
115	C1 Voltage	1mV (Unsigned Integer) (Value * 1000)	55,50V

¹ Number of decimals depend on the rounding of received Value. Micro calculation is: (40550*8)/65536

b. Reply after calibration

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFyyx	0x00	0x20	CalPtr	Result	New Calibration Value (MSB)			

Result:

Result	Status
-2	Incorrect CalPtr value
-1	Calibration value out of range
0	Unable to save to EEPROM
1	Calibration success

New Calibration Value: The format is a 32bit value fixed point where the upper 16bit is value and lower 16bit is fragment.

9.2.8 App Settings transfer (BlockID, ID)

Settings in the Li-G4 can be adjusted via CAN communication by BlockID and ID. realtime values and internal functions is included in setting transfer and can be address via reserved BlockID's.

BlockID #	Data Type	Description
0	Real Time values	See chapter: 5.1.1
1	Functions	Not available
2-199	EEPROM Settings	See chapter: 7

App Settings Request Value

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x40	BlockID	ID	Value			

- BlockID points to the settings block. (See settings Table)
- ID points to the specific setting in settings block. (See settings Table)
- Value read from settings (Value is fixed point signed long, LSB first)
bit 0-15 = Fraction.
bit 16-31 = Decimal.

App Settings Save Value

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x41	BlockID	ID	Value			

- BlockID points to the settings block. (See settings Table)
- ID points to the specific setting in settings block. (See settings Table)
- Value saved to settings (Value is fixed point signed long, LSB first)
bit 0-15 = Fraction.
bit 16-31 = Decimal.
- After settings has been saved is as similar message returned with the save value.

App Settings Request Default, Min, Max

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x42-0x44	BlockID	ID	Value			

- Data2 - Command:
0x42 = Default value
0x43 = Min Value
0x44 = Max Value
- BlockID points to the settings block. (See settings Table)
- ID points to the specific setting in settings block. (See settings Table)
- Value read from settings (Value is fixed point signed long, LSB first)
bit 0-15 = Fraction.
bit 16-31 = Decimal.